

Linear-Log Regression Analysis

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Learning Objectives

- Determine the appropriate ICD codes, histologic codes, and any relevant subtypes of a cancer.
- Utilize SEER*Stat to access data from the SEER cancer registry.
- Visualize cancer incidence and trends by sex, race, and cancer subtype.

Code

```
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ dplyr 1.1.4 ✓ readr 2.1.5

✓ forcats 1.0.0 ✓ stringr 1.5.1

✓ lubridate 1.9.4 ✓ tibble 3.3.0

✓ purrr 1.1.0 ✓ tidyr 1.3.1

— Conflicts — tidyverse_conflicts() —

✗ dplyr::filter() masks stats::filter()

✗ dplyr::lag() masks stats::lag()

ℹ Use the conflicted package (<<http://conflicted.r-lib.org/>>) to force all conflicts to become errors

```
library(ggplot2)
```

```
library(dplyr)
```

```
data <- read.csv("Melanoma.csv", header=TRUE)
```

```
colnames(data) <- c("Race", "Sex", "Subsite", "Year", "Rate", "Cases", "Pop")
```

```
# Remove the year range
```

```
data_melanoma <- data |> filter(Year != "1975-2022")
```

```
data_melanoma$Rate = as.numeric(as.character(data_melanoma$Rate))
```

Warning: NAs introduced by coercion

```
data_melanoma$Year = as.numeric(as.character(data_melanoma$Year))
```

```
# Example: Subset to the overall
```

```
data_melanoma_overall <- data_melanoma %>%
```

```
  filter(
```

```
    Race == "All races",
```

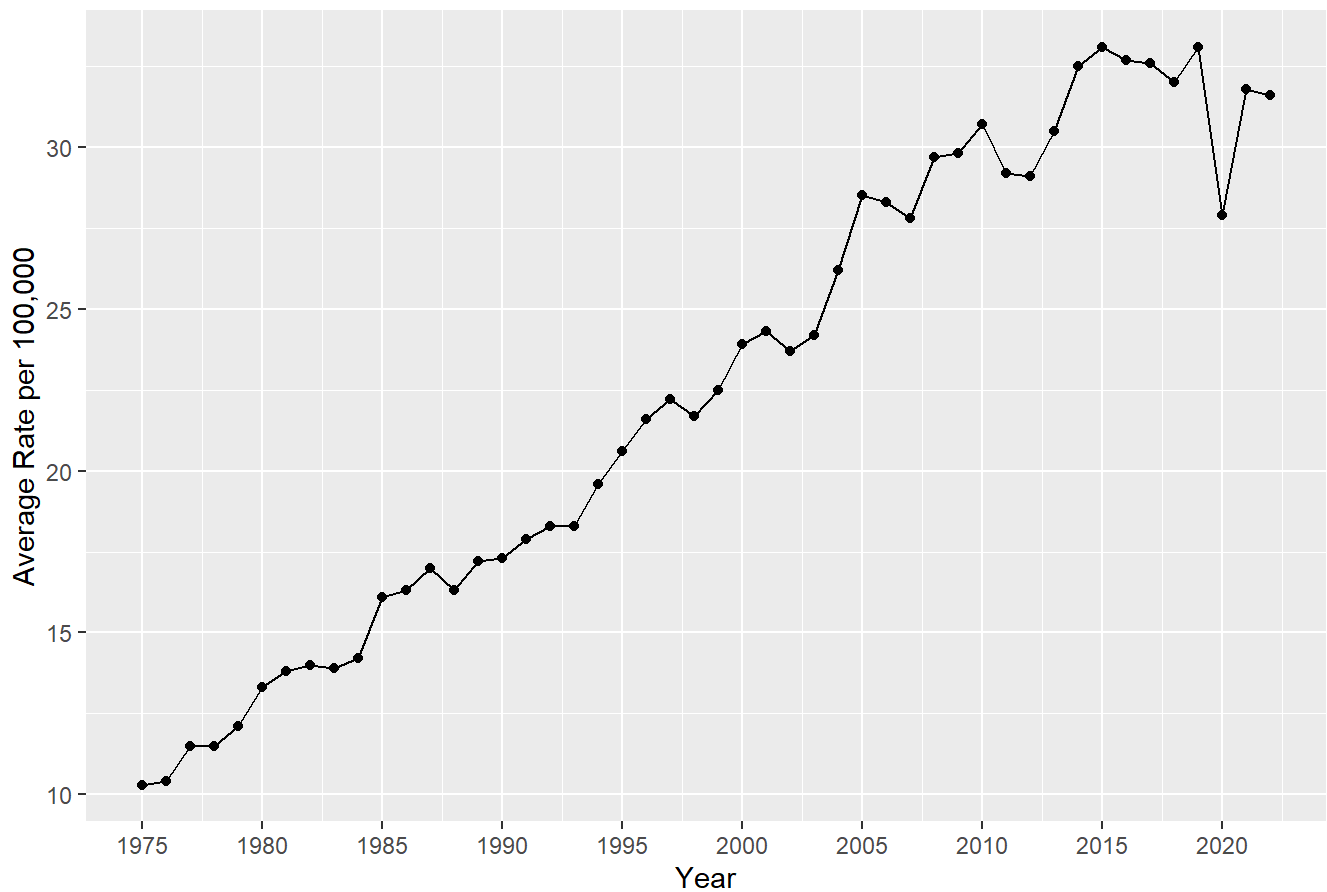
```
    Sex == "Male and female",
```

```
Subsite %in% c(
  "Head and Neck Anatomic site of Melanoma",
  "Trunk Anatomic site of Melanoma",
  "Arm (shoulder to finger) Anatomic site of Melanoma",
  "Leg (hip to toe) Anatomic site of Melanoma")) %>%
dplyr::select(Year, Rate, Cases, Pop)

data_melanoma_overall_combined = data_melanoma_overall %>%
  group_by(Year) %>%
  summarise(
    total_cases = sum(Cases, na.rm = TRUE),
    total_rate = sum(Rate, na.rm = TRUE),
    .groups = "drop")

ggplot(data_melanoma_overall_combined, aes(x = Year, y = total_rate)) +
  geom_line() +
  geom_point() +
  labs(title = "Average Yearly Melanoma Age Adjusted Rate from 1975 to 2022 ",
       x = "Year",
       y = "Average Rate per 100,000") +
  scale_x_continuous(breaks = seq(1975, 2025, by = 5))
```

Average Yearly Melanoma Age Adjusted Rate from 1975 to 2022



```
# Age-adjusted incidence rates by sex

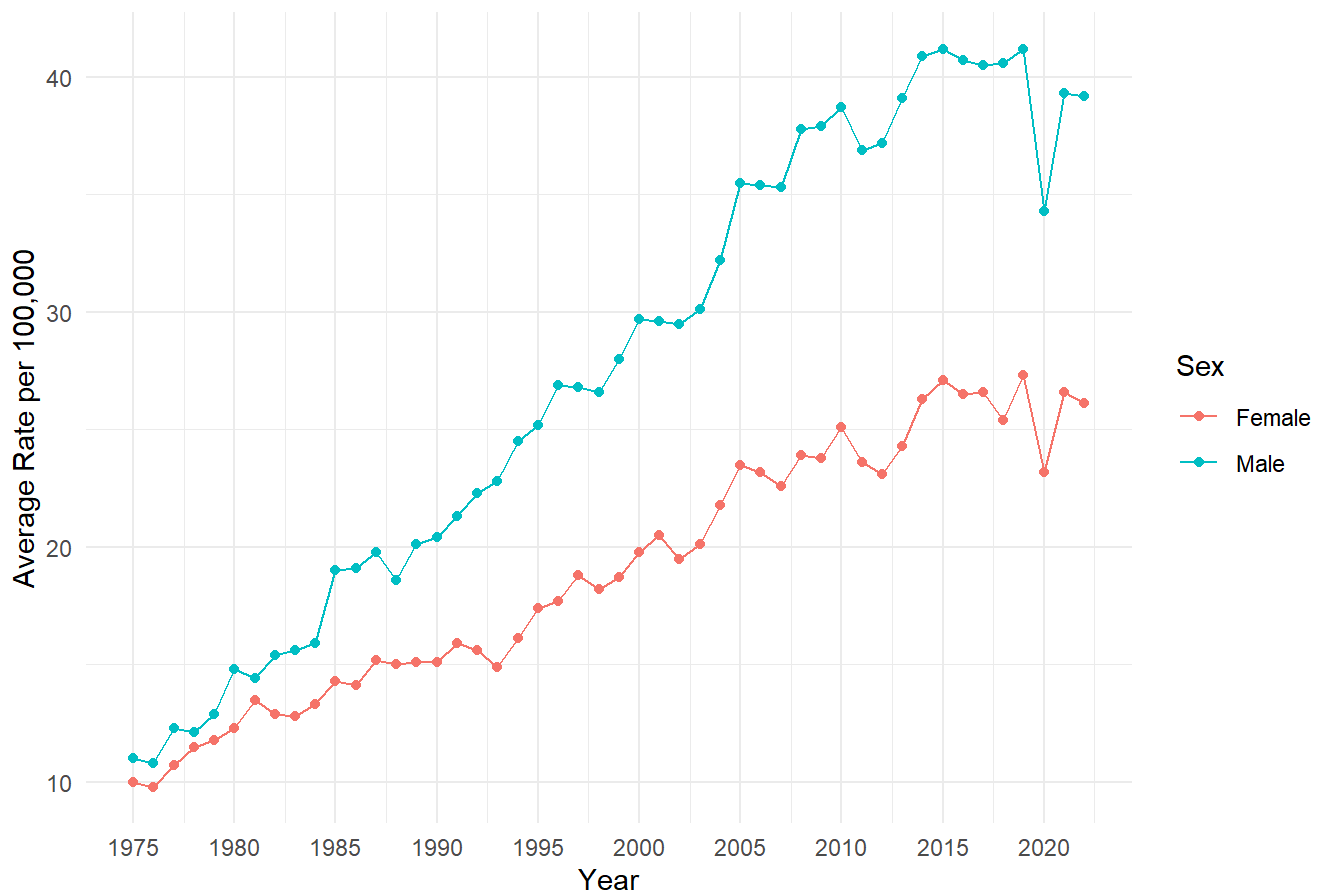
data_melanoma$Rate <- as.numeric(as.character(data_melanoma$Rate))
data_melanoma$Year <- as.numeric(as.character(data_melanoma$Year))

# Filter for relevant subsites and races, but keep Sex as a variable
data_melanoma_by_sex <- data_melanoma |>
  filter(
    Race == "All races",
    Sex %in% c("Male", "Female"),
    Subsite %in% c(
      "Head and Neck Anatomic site of Melanoma",
      "Trunk Anatomic site of Melanoma",
      "Arm (shoulder to finger) Anatomic site of Melanoma",
      "Leg (hip to toe) Anatomic site of Melanoma" )) %>%
  dplyr::select(Year, Rate, Sex, Cases, Pop)

# Group and summarize by Year and Sex
data_melanoma_by_sex_summary <- data_melanoma_by_sex %>%
  group_by(Year, Sex) %>%
  summarise(
    total_cases = sum(Cases, na.rm = TRUE),
    total_rate = sum(Rate, na.rm = TRUE),
    .groups = "drop")

# Plot age-adjusted incidence rates by sex
ggplot(data_melanoma_by_sex_summary, aes(x = Year, y = total_rate, color = Sex)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Age-Adjusted Melanoma Incidence Rates by Sex (1975-2022)",
    x = "Year",
    y = "Average Rate per 100,000", color = "Sex") +
  scale_x_continuous(breaks = seq(1975, 2025, by = 5)) +
  theme_minimal()
```

Age-Adjusted Melanoma Incidence Rates by Sex (1975–2022)



```
# Age-adjusted incidence rates by race

# Convert columns to numeric
data_melanoma$Rate <- as.numeric(as.character(data_melanoma$Rate))
data_melanoma$Year <- as.numeric(as.character(data_melanoma$Year))

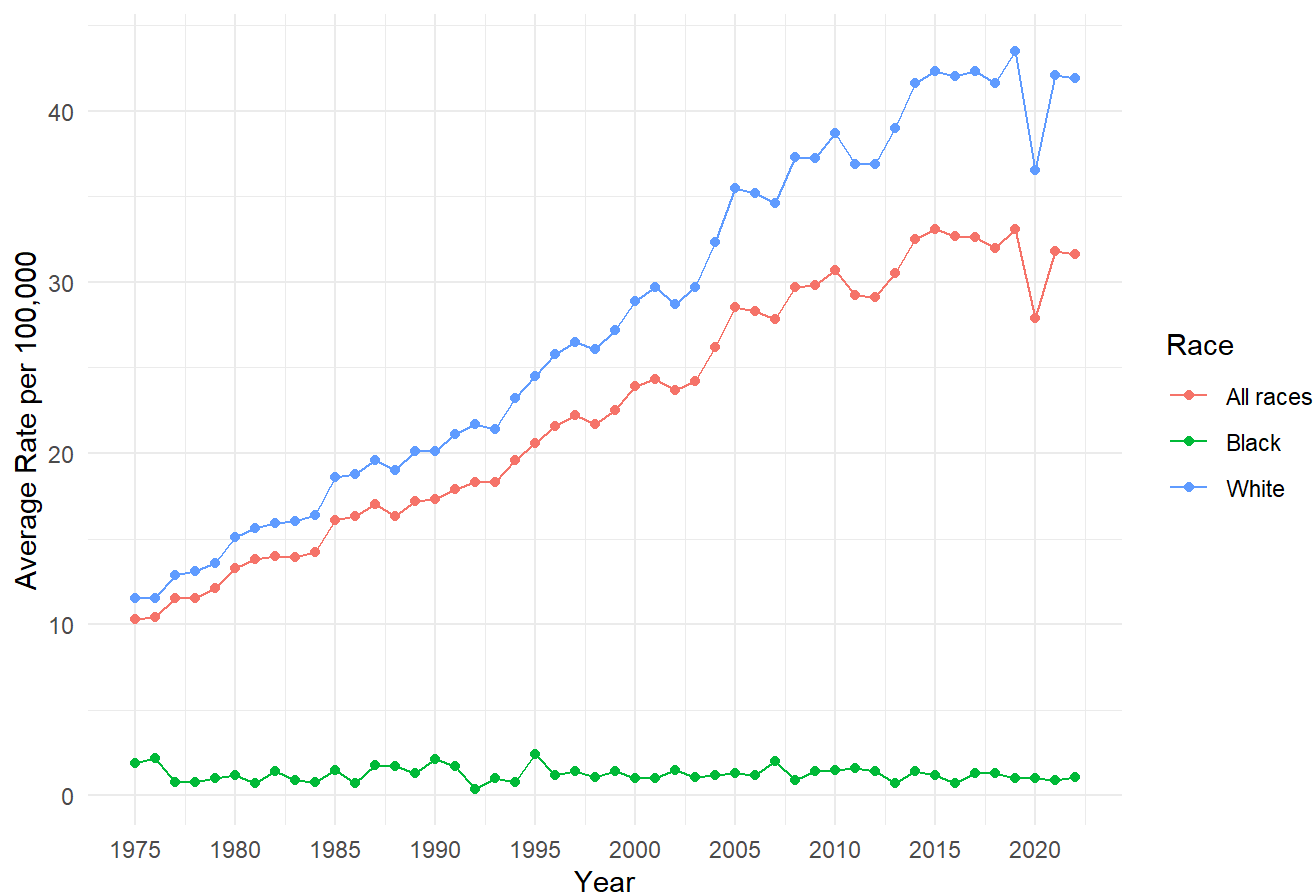
# Filter for relevant subsites and sexes, but keep Race as a variable
data_melanoma_by_race <- data_melanoma |>
  filter(
    Sex == "Male and female",
    Race %in% c("White", "Black", "All races"),
    Subsite %in% c(
      "Head and Neck Anatomic site of Melanoma",
      "Trunk Anatomic site of Melanoma",
      "Arm (shoulder to finger) Anatomic site of Melanoma",
      "Leg (hip to toe) Anatomic site of Melanoma" )) %>%
  dplyr::select(Year, Rate, Race, Cases, Pop)

# Group and summarize by year and race
data_melanoma_by_race_summary <- data_melanoma_by_race |>
  group_by(Year, Race) |>
  summarise(
    total_cases = sum(Cases, na.rm = TRUE),
```

```
total_rate = sum(Rate, na.rm = TRUE),
.groups = "drop")

# Plot age-adjusted incidence rates by race
ggplot(data_melanoma_by_race_summary, aes(x = Year, y = total_rate, color = Race)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Age-Adjusted Melanoma Incidence Rates by Race (1975-2022)",
    x = "Year",
    y = "Average Rate per 100,000", color = "Race" ) +
  scale_x_continuous(breaks = seq(1975, 2025, by = 5)) +
  theme_minimal()
```

Age-Adjusted Melanoma Incidence Rates by Race (1975–2022)



```
# Age-adjusted incidence rates by subsite

# Convert columns to numeric
data_melanoma$Rate <- as.numeric(as.character(data_melanoma$Rate))
data_melanoma$Year <- as.numeric(as.character(data_melanoma$Year))

# Filter for relevant races and sexes, but keep subsite as a variable
data_melanoma_by_subsite <- data_melanoma |>
  filter(
```

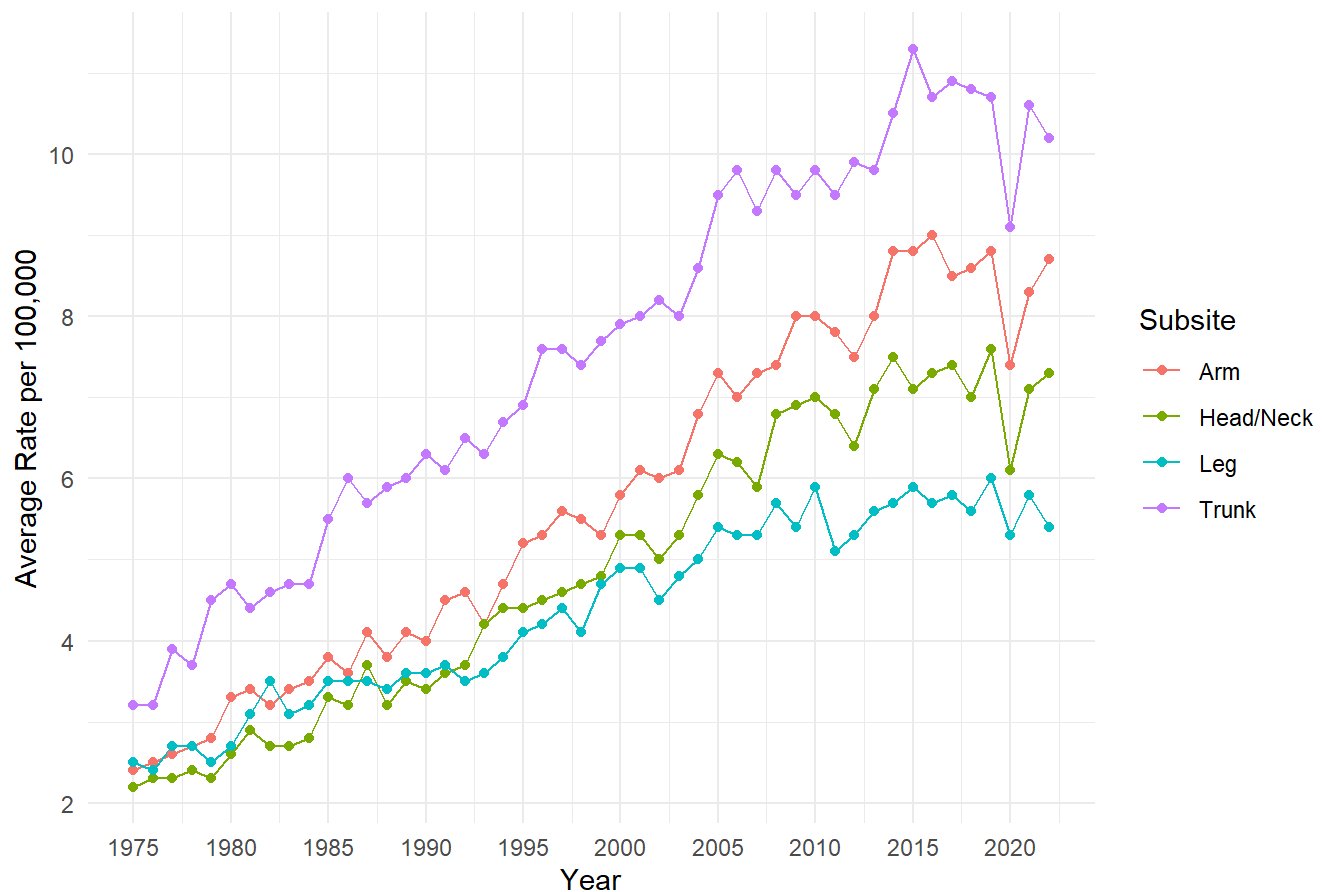
```
Race == "All races",
Sex == "Male and female",
Subsite %in% c(
  "Head and Neck Anatomic site of Melanoma",
  "Trunk Anatomic site of Melanoma",
  "Arm (shoulder to finger) Anatomic site of Melanoma",
  "Leg (hip to toe) Anatomic site of Melanoma")) %>%
dplyr::select(Year, Rate, Subsite, Cases, Pop)

# Rename subsite to a more condensed form for nicer plotting legend
data_melanoma_by_subsite <- data_melanoma_by_subsite |>
mutate(Subsite = recode(Subsite,
  "Head and Neck Anatomic site of Melanoma" = "Head/Neck",
  "Trunk Anatomic site of Melanoma" = "Trunk",
  "Arm (shoulder to finger) Anatomic site of Melanoma" = "Arm",
  "Leg (hip to toe) Anatomic site of Melanoma" = "Leg"))

# Group and summarize by year and subsite
data_melanoma_by_subsite_summary <- data_melanoma_by_subsite |>
group_by(Year, Subsite) |>
summarise(
  total_cases = sum(Cases, na.rm = TRUE),
  total_rate = mean(Rate, na.rm = TRUE),
  .groups = "drop")

# Plot age-adjusted incidence rates by subsite
ggplot(data_melanoma_by_subsite_summary, aes(x = Year, y = total_rate, color = Subsite)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Age-Adjusted Melanoma Incidence Rates by Subsite (1975-2022)",
    x = "Year",
    y = "Average Rate per 100,000",
    color = "Subsite") +
  scale_x_continuous(breaks = seq(1975, 2025, by = 5)) +
  theme_minimal()
```

Age-Adjusted Melanoma Incidence Rates by Subsite (1975–2022)



```
# Age-adjusted incidence rates by race and sex

# Filter and clean
data_melanoma$Rate <- as.numeric(as.character(data_melanoma$Rate))
data_melanoma$Year <- as.numeric(as.character(data_melanoma$Year))

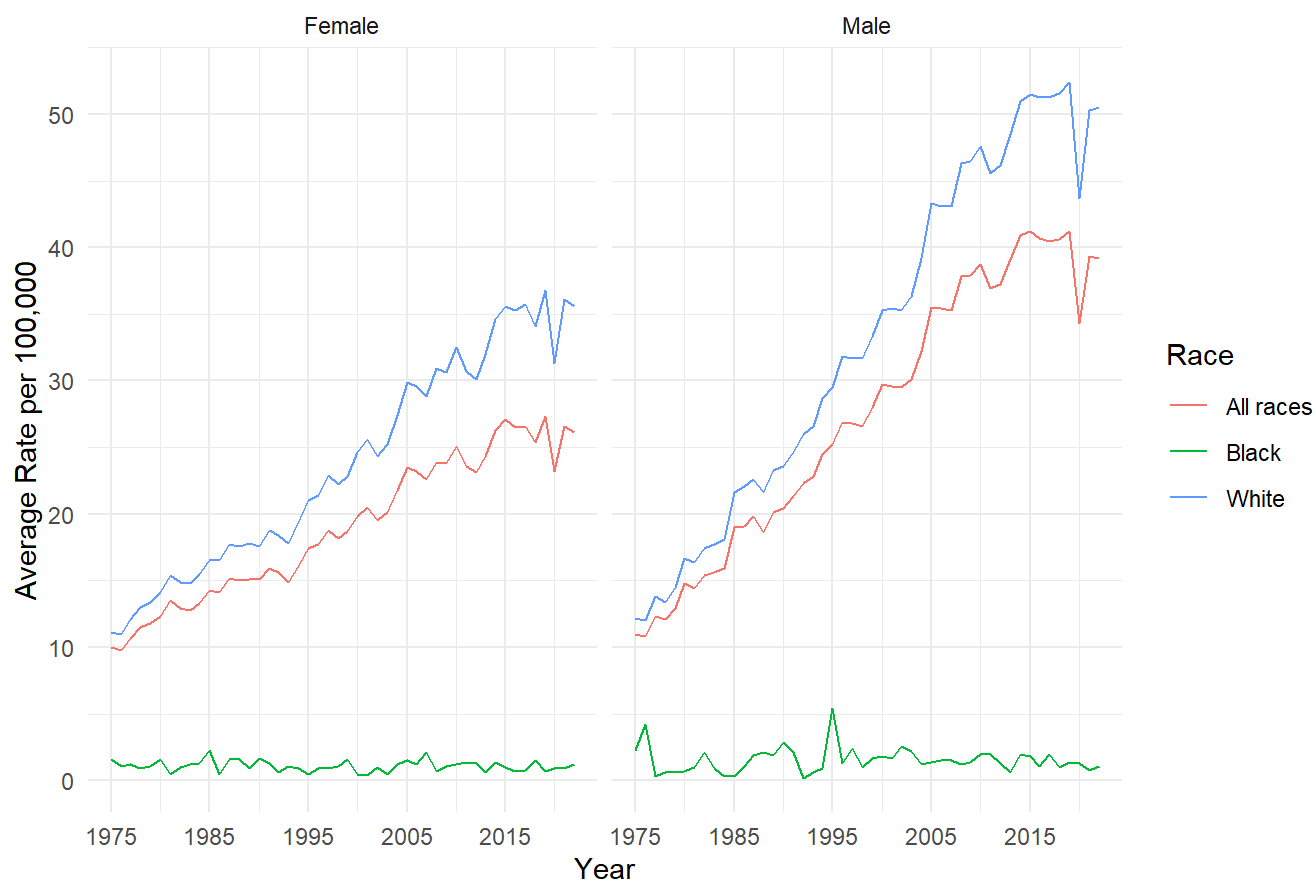
data_melanoma_race_sex <- data_melanoma |>
  filter(
    Race %in% c("White", "Black", "All races"),
    Sex %in% c("Male", "Female"),
    Subsite %in% c(
      "Head and Neck Anatomic site of Melanoma",
      "Trunk Anatomic site of Melanoma",
      "Arm (shoulder to finger) Anatomic site of Melanoma",
      "Leg (hip to toe) Anatomic site of Melanoma")) %>%
  dplyr::select(Year, Rate, Race, Sex, Cases, Pop)

# Group and summarize by year and subsite
data_melanoma_race_sex_summary <- data_melanoma_race_sex |>
  group_by(Year, Race, Sex) |>
  summarise(
    total_cases = sum(Cases, na.rm = TRUE),
```

```
total_rate = sum(Rate, na.rm = TRUE),
.groups = "drop")

# Plot
ggplot(data_melanoma_race_sex_summary, aes(x = Year, y = total_rate, color = Race)) +
  geom_line() +
  facet_wrap(~Sex) +
  labs(
    title = "Melanoma Incidence by Race and Sex (1975–2022)",
    x = "Year",
    y = "Average Rate per 100,000", color = "Race") +
  scale_x_continuous(breaks = seq(1975, 2025, by = 10)) +
  theme_minimal()
```

Melanoma Incidence by Race and Sex (1975–2022)



```
# Age-adjusted incidence rates by race and subsite

data_melanoma$Rate <- as.numeric(as.character(data_melanoma$Rate))
data_melanoma$Year <- as.numeric(as.character(data_melanoma$Year))

data_melanoma_race_subsite <- data_melanoma |>
  filter(
    Race %in% c("White", "Black", "All races"),
    Sex == "Male and female",
```

```
  Subsite %in% c(
    "Head and Neck Anatomic site of Melanoma",
    "Trunk Anatomic site of Melanoma",
    "Arm (shoulder to finger) Anatomic site of Melanoma",
    "Leg (hip to toe) Anatomic site of Melanoma")) %>%
dplyr::select(Year, Rate, Race, Subsite, Cases, Pop)

# Rename subsite to a more condensed form for nicer plotting legend
data_melanoma_race_subsite <- data_melanoma_race_subsite |>
  mutate(Subsite = recode(Subsite,
    "Head and Neck Anatomic site of Melanoma" = "Head/Neck",
    "Trunk Anatomic site of Melanoma" = "Trunk",
    "Arm (shoulder to finger) Anatomic site of Melanoma" = "Arm",
    "Leg (hip to toe) Anatomic site of Melanoma" = "Leg"))

# Group and summarize by year and subsite
data_melanoma_race_subsite_summary <- data_melanoma_race_subsite |>
  group_by(Year, Race, Subsite) |>
  summarise(
    total_cases = sum(Cases, na.rm = TRUE),
    total_rate = sum(Rate, na.rm = TRUE),
    .groups = "drop")

# Plot incidence by race and subsite
ggplot(data_melanoma_race_subsite_summary, aes(x = Year, y = total_rate, color = Race)) +
  geom_line() +
  facet_wrap(~Subsite) +
  labs(
    title = "Melanoma Incidence by Race and Subsite (1975-2022)",
    x = "Year",
    y = "Average Rate per 100,000", color = "Race") +
  scale_x_continuous(breaks = seq(1975, 2025, by = 10)) +
  theme_minimal()
```

Melanoma Incidence by Race and Subsite (1975–2022)

